

release air pollution that constitutes about one-fourth of the global black-carbon emissions that warm the planet and kill roughly 4 million people a year. Clean cookstoves simultaneously mitigate climate change and indoor air pollution-related mortality. Of note, evidence has shown that long-term air pollution exposure may contribute to mortality risk from COVID-19.

Climate *adaptation* is secondary prevention and seeks to reduce harms associated with sea level rise, heat waves, floods, droughts, wildfires, and other greenhouse gas-driven events. The efficacy of adaptation is constrained by the challenges inherent in predicting the precise location, duration, and severity of extreme weather events and flooding related to sea level rise, among other considerations.

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category has not reliably predicted infection with resistant pathogens and has been associated with increased use of broad-spectrum antibiotics, particularly those employed for treatment of methicillin-resistant *Staphylococcus aureus* (MRSA) and antipseudomonal β -lactams. Accordingly, use of the HCAP category should be discontinued. Rather than relying on a predefined subset of pneumonia cases, it is better to assess patients individually on the basis of risk factors for infection with a resistant organism. Risk factors for infection with MRSA and *Pseudomonas aeruginosa* include prior isolation of the organism, particularly from the respiratory tract during the preceding year, and/or hospitalization and treatment with an antibiotic in the previous 90 days.

Pneumonia caused by macroaspiration of oropharyngeal or gastric contents, usually referred to as aspiration pneumonia, is best thought of as a point on the continuum that includes CAP and HAP. Estimates suggest that aspiration pneumonia accounts for 5–15% of CAP cases, but reliable figures for HAP are unavailable. The airways or pulmonary parenchyma may be involved, and patients usually represent a clinical phenotype with risk factors for macroaspiration and involvement of characteristic anatomic pulmonary locations.

PATHOPHYSIOLOGY

Pneumonia is the result of the proliferation of microbial pathogens at the alveolar level and the host's response to them. Until recently, it was thought that the lungs were sterile and that pneumonia resulted from the introduction of potential pathogens into this sterile environment. Typically, this introduction occurred through microaspiration of oropharyngeal organisms into the lower respiratory tract. Overcoming of innate and adaptive immunity by such microorganisms could result in the clinical syndrome of pneumonia.

Recent use of culture-independent techniques of microbial identification has demonstrated a complex and diverse community of bacteria in the lungs that constitutes the lung microbiota. Awareness of this microbiota has prompted a rethinking of how pneumonia develops. Mechanical factors, such as the hairs and turbinates of the nares, the branching tracheobronchial tree, mucociliary clearance, and gag and cough reflexes, all play a role in host defense but are insufficient to effectively block bacterial access to the lower airways. In the absence of a sufficient barrier, microorganisms may reach the lower respiratory tract by a variety of pathways, including inhalation, microaspiration, and direct mucosal dispersion.

The constitution of the lung microbiota is determined by three factors: microbial entry into the lungs, microbial elimination, and regional growth conditions for bacteria, such as pH, oxygen tension, and temperature. The key question, however, is how a dynamic homeostasis among bacterial communities results in acute infection. Pneumonia therefore does not appear to be the result of the invasion of a sterile space by a particular microorganism but is more likely an emergent phenomenon dependent upon a number of mechanisms, including self-accelerating positive feedback loops.

A possible model for pneumonia is as follows. An inflammatory event resulting in epithelial and/or endothelial injury results in the release of cytokines, chemokines, and catecholamines, some of which may selectively promote the growth of certain bacteria, such as *Streptococcus pneumoniae* and *P. aeruginosa*. This cycle of inflammation, enhanced nutrient availability, and release of potential bacterial growth factors may result in a positive feedback loop that further accelerates inflammation and the growth of particular bacteria, which may then become dominant. In cases of CAP and HAP, the trigger may be a viral infection compounded by microaspiration of oropharyngeal organisms. In cases of true aspiration pneumonia, the trigger may simply be the macroaspiration event itself.

Once triggered, innate and adaptive immune responses can ideally help contain potential pathogens and prevent the development of pneumonia. However, in the face of continuing inflammation (and especially if a positive feedback loop becomes sustainable), the process may proceed to a full-fledged pneumonia syndrome. Inflammatory mediators such as interleukin 6 and tumor necrosis factor result in fever, and chemokines such as interleukin 8 and granulocyte

Section 2 Clinical Syndromes: Community-Acquired Infections

126 Pneumonia

Lionel A. Mandell, Michael S. Niederman

DEFINITION

Pneumonia is an infection of the pulmonary parenchyma. Despite significant morbidity and mortality, it is often misdiagnosed, mistreated, and underestimated. Pneumonia has usually been classified as community-acquired (CAP), hospital-acquired (HAP), or ventilator-associated (VAP). A fourth category, health care-associated pneumonia (HCAP) was introduced to encompass cases caused by multidrug-resistant (MDR) pathogens typically associated with HAP and cases in unhospitalized individuals at risk of MDR infection. Unfortunately, this

1010 colony-stimulating factor increase local neutrophil numbers. Mediators released by macrophages and neutrophils may create an alveolar capillary leak resulting in impaired oxygenation, hypoxemia, and radiographic infiltrates. Moreover, some bacterial pathogens appear to interfere with the hypoxic vasoconstriction that would normally occur with fluid-filled alveoli, and this interference may result in severe hypoxemia. Decreased compliance due to capillary leak, hypoxemia, increased respiratory drive, increased secretions, and occasionally infection-related bronchospasm all lead to worsening dyspnea. If severe enough, changes in lung mechanics secondary to reductions in lung volume, compliance, and intrapulmonary shunting of blood may cause respiratory failure.

Cardiovascular events with pneumonia, particularly in the elderly and usually in association with pneumococcal pneumonia and influenza, are increasingly recognized. These events, which may be acute or whose occurrence may extend to at least 1 year, include congestive heart failure, arrhythmia, myocardial infarction, or stroke and may be caused by a variety of mechanisms, including increased myocardial load and/or destabilization of atherosclerotic plaques by inflammation. In animal models, direct myocardial invasion by pneumococci may result in scarring and impaired myocardial function and conductivity.

PATHOLOGY

Classic pneumonia evolves through a series of stages. The initial stage is edema with a proteinaceous exudate and often bacteria in the alveoli. Next is a rapid transition to the red hepatization phase. Erythrocytes in the intraalveolar exudate give this stage its name. In the third phase, gray hepatization, no new erythrocytes are extravasating, and those already present have been lysed and degraded. The neutrophil is the predominant cell, fibrin deposition is abundant, and bacteria have disappeared. This phase corresponds with the successful containment of the infection and improvement in gas exchange. In the final phase, resolution, the macrophage reappears as the dominant cell in the alveolar space and the debris of neutrophils, and bacteria and fibrin have been cleared, as has the inflammatory response.

This pattern has been described best for lobar pneumococcal pneumonia but may not apply to pneumonia of all etiologies. In VAP, respiratory bronchiolitis may precede the development of a radiologically apparent infiltrate. A bronchopneumonia pattern is most common in nosocomial pneumonias, whereas a lobar pattern is more common in bacterial CAP. Despite the radiographic appearance, viral and *Pneumocystis* pneumonias represent alveolar rather than interstitial processes.

COMMUNITY-ACQUIRED PNEUMONIA

ETIOLOGY

The list of potential etiologic agents of CAP includes bacteria, fungi, viruses, and protozoa. Newer viral pathogens include metapneumoviruses, the coronaviruses responsible for severe acute respiratory syndrome (SARS) and Middle East respiratory syndrome (MERS), and the recently discovered coronavirus that originated in Wuhan, China, and is designated SARS-CoV-2. First described in December 2019, SARS-CoV-2 and its associated clinical disease, COVID-19, have reached pandemic proportions and are a cause of significant morbidity and mortality. The virus and the disease are discussed in detail in [Chap. 199](#).

Although most CAP cases are caused by relatively few pathogens, an accurate determination of their prevalence is difficult because laboratory testing methods are often insensitive and indirect ([Table 126-1](#)). Separation of potential agents into “typical” bacterial pathogens and “atypical” organisms may be helpful. The former group includes *S. pneumoniae*, *Haemophilus influenzae*, and, in selected patients, *S. aureus* and gram-negative bacilli such as *Klebsiella pneumoniae* and *P. aeruginosa*. The “atypical” organisms include *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, and *Legionella* species as well as respiratory viruses such as influenza virus, adenoviruses, human metapneumoviruses, respiratory syncytial virus, and coronaviruses. With the increasing use of pneumococcal vaccine, the incidence of pneumococcal pneumonia is decreasing. Cases due to *M. pneumoniae* and *C. pneumoniae*, however,

TABLE 126-1 Microbial Causes of Community-Acquired Pneumonia, by Site of Care

OUTPATIENTS	HOSPITALIZED PATIENTS	
	NON-ICU	ICU
<i>Streptococcus pneumoniae</i>	<i>S. pneumoniae</i>	<i>S. pneumoniae</i>
<i>Mycoplasma pneumoniae</i>	<i>M. pneumoniae</i>	<i>Staphylococcus aureus</i>
<i>Haemophilus influenzae</i>	<i>Chlamydia pneumoniae</i>	<i>Legionella</i> spp.
<i>C. pneumoniae</i>	<i>H. influenzae</i>	Gram-negative bacilli
Respiratory viruses ^a	<i>Legionella</i> spp.	<i>H. influenzae</i>
	Respiratory viruses ^a	Respiratory viruses

^aInfluenza A and B viruses, human metapneumovirus, adenoviruses, respiratory syncytial viruses, parainfluenza viruses, coronaviruses (e.g., SARS-CoV-2).

Abbreviation: ICU, intensive care unit.

appear to be increasing, especially among young adults. Viruses are recognized as increasingly important in pneumonia, and polymerase chain reaction (PCR)-based testing indicates their presence in the respiratory tract of 20–30% of healthy adults and in the same percentage of pneumonia patients, including those who are severely ill. The most common are influenza, parainfluenza, and respiratory syncytial viruses. Whether they are true etiologic pathogens, co-pathogens, or simply colonizers cannot always be determined. Atypical organisms cannot be cultured on standard media or seen on Gram's stain, but their frequency and importance have significant implications for therapy. They are intrinsically resistant to all β -lactams and require treatment with a macrolide, a fluoroquinolone, or a tetracycline. In the 10–15% of CAP cases that are polymicrobial, the etiology usually includes a combination of typical and atypical pathogens.

Earlier literature suggested that aspiration pneumonia was caused primarily by anaerobes, with or without aerobic pathogens. A shift, however, has been noted recently: if aspiration pneumonia is acquired in a community or hospital setting, the likely pathogens are those usually associated with CAP or HAP. Anaerobes may still play a role, especially in patients with poor dentition, lung abscess, necrotizing pneumonia, or empyema.

S. aureus pneumonia is known to complicate influenza virus infection. However, MRSA has been reported as a primary etiologic agent of CAP. Although cases caused by MRSA are relatively uncommon, clinicians must be aware of its potentially serious consequences, such as necrotizing pneumonia. Two factors have led to this problem: the spread of MRSA from the hospital setting to the community and the emergence of genetically distinct strains of MRSA in the community. Community-associated MRSA (CA-MRSA) strains may infect healthy individuals who have had no association with health care.

Despite a careful history, physical examination, and radiographic studies, the causative pathogen is often difficult to predict with certainty, and in more than half of cases a specific etiology is not determined. Nevertheless, epidemiologic and risk factors may suggest certain pathogens ([Table 126-2](#)).

EPIDEMIOLOGY

More than 5 million CAP cases occur annually in the United States. Along with influenza, CAP is the eighth leading cause of death in this country. CAP causes more than 55,000 deaths annually and results in more than 1.2 million hospitalizations; ~70% of patients are treated as outpatients and 30% as inpatients. The mortality rate among outpatients is usually <5% but ranges from ~12% to 40% among hospitalized patients, with the exact rate depending on whether treatment takes place in or outside the intensive care unit (ICU). In the United States, CAP is the leading cause of death from infection among patients >65 years of age. Moreover, 18% of hospitalized CAP patients are readmitted within 1 month of discharge. The overall yearly CAP cost is estimated at \$17 billion. The overall incidence among adults is ~16–23 cases per 1000 persons per year, with the highest rates at the extremes of age.

The risk factors for CAP in general and for pneumococcal pneumonia in particular have implications for treatment. They include

TABLE 126-2 Epidemiologic Factors Suggesting Possible Causes of Community-Acquired Pneumonia	
FACTOR	POSSIBLE PATHOGEN(S)
Alcoholism	<i>Streptococcus pneumoniae</i> , oral anaerobes, <i>Klebsiella pneumoniae</i> , <i>Acinetobacter</i> spp., <i>Mycobacterium tuberculosis</i>
COPD and/or smoking	<i>Haemophilus influenzae</i> , <i>Pseudomonas aeruginosa</i> , <i>Legionella</i> spp., <i>S. pneumoniae</i> , <i>Moraxella catarrhalis</i> , <i>Chlamydia pneumoniae</i>
Structural lung disease (e.g., bronchiectasis)	<i>P. aeruginosa</i> , <i>Burkholderia cepacia</i> , <i>Staphylococcus aureus</i>
Dementia, stroke, decreased level of consciousness	Oral anaerobes, gram-negative enteric bacteria
Lung abscess	CA-MRSA, oral anaerobes, endemic fungi, <i>M. tuberculosis</i> , atypical mycobacteria
Travel to Ohio or St. Lawrence river valley	<i>Histoplasma capsulatum</i>
Travel to southwestern United States	Hantavirus, <i>Coccidioides</i> spp.
Travel to Southeast Asia	<i>Burkholderia pseudomallei</i> , avian influenza virus
Stay in hotel or on cruise ship in previous 2 weeks	<i>Legionella</i> spp.
Local influenza activity	Influenza virus, <i>S. pneumoniae</i> , <i>S. aureus</i>
Exposure to infected humans	SARS-CoV-2
Exposure to birds	<i>H. capsulatum</i> , <i>Chlamydia psittaci</i>
Exposure to rabbits	<i>Francisella tularensis</i>
Exposure to sheep, goats, parturient cats	<i>Coxiella burnetii</i>

Abbreviations: CA-MRSA, community-acquired methicillin-resistant *Staphylococcus aureus*; COPD, chronic obstructive pulmonary disease; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

alcoholism, asthma, immunosuppression, institutionalization, and age >70 years. In the elderly, decreased cough and gag reflexes and reduced antibody and Toll-like receptor responses increase the likelihood of pneumonia. Risk factors for pneumococcal pneumonia include dementia, seizure disorders, heart failure, cerebrovascular disease, alcoholism, tobacco smoking, chronic obstructive pulmonary disease (COPD), and HIV infection. CA-MRSA pneumonia is more likely in patients with skin colonization or infection with CA-MRSA and after viral infection. Enterobacteriaceae tend to infect patients who have recently been hospitalized or given antibiotics or who have comorbidities such as alcoholism, heart failure, or renal failure. *P. aeruginosa* is a particular problem in patients with severe structural lung disease (e.g., bronchiectasis, cystic fibrosis, or severe COPD). Risk factors for *Legionella* infection include diabetes, hematologic malignancy, cancer, severe renal disease, HIV infection, smoking, male gender, and a recent hotel stay or trip on a cruise ship.

CLINICAL MANIFESTATIONS

The clinical presentation of pneumonia can vary from indolent to fulminant and from mild to fatal in severity. Manifestations of worsening severity include both constitutional findings and those limited to the lung and associated structures. The patient is frequently febrile and/or tachycardic and may experience chills and/or sweats. Cough may be nonproductive or productive of mucoid, purulent, or blood-tinged sputum. Gross hemoptysis is suggestive of necrotizing pneumonia (e.g., that due to CA-MRSA). Depending on severity, the patient may be able to speak in full sentences or may be short of breath. With pleural involvement, the patient may experience pleuritic chest pain. Up to 20% of patients may have gastrointestinal symptoms such as nausea, vomiting, or diarrhea. Other symptoms may include fatigue, headache, myalgias, and arthralgias.

Findings on physical examination vary with the degree of pulmonary consolidation and the presence or absence of a significant pleural

effusion. An increased respiratory rate and use of accessory muscles of respiration are common. Palpation may reveal increased or decreased tactile fremitus, and the percussion note can vary from dull to flat, reflecting underlying consolidated lung and pleural fluid, respectively. Crackles, bronchial breath sounds, and possibly a pleural friction rub may be heard. The clinical presentation may be less obvious in the elderly, who may initially display new-onset or worsening confusion but few other manifestations. Severely ill patients may have septic shock and evidence of organ failure. In cases of CAP, symptoms can range from almost nonexistent to severe, and chest radiographic findings are often in gravity-dependent parts of the lung.

DIAGNOSIS

When confronted with possible CAP, the physician must ask two questions: is this pneumonia, and, if so, what is the likely pathogen? The former question is answered by clinical and radiographic methods, whereas the latter requires laboratory techniques.

Clinical Diagnosis The differential diagnosis includes infectious and noninfectious entities, including acute bronchitis, exacerbations of chronic bronchitis, heart failure, and pulmonary embolism. The importance of a careful history cannot be overemphasized. The diagnosis of CAP requires a compatible history, such as cough, sputum production, fever and dyspnea, and a new infiltrate on chest radiography.

Unfortunately, the sensitivity and specificity of findings on physical examination are only 58% and 67%, respectively. Chest radiography is often necessary to differentiate CAP from other conditions. Radiographic findings may suggest increased severity (e.g., cavitation or multilobar involvement). Occasionally, radiographic results suggest an etiologic diagnosis, such as pneumatoceles in *S. aureus* infection or an upper-lobe cavitating lesion in tuberculosis. CT may be of value in suspected loculated effusion or cavity cases or in postobstructive pneumonia caused by a tumor or foreign body. For outpatients, clinical and radiologic assessments are usually all that is required before treatment is started since most laboratory results are not available soon enough to influence initial management. In certain cases, the availability of rapid point-of-care outpatient tests can be important; for example, rapid diagnosis of influenza infection can prompt specific anti-influenza treatment and secondary prevention measures.

Etiologic Diagnosis The etiology of pneumonia usually cannot be determined solely on the basis of clinical or radiographic presentation. Data from more than 17,000 emergency department CAP cases showed an etiologic determination in only 7.6%. Except for CAP patients admitted to the ICU, no data exist to show that treatment directed at a specific pathogen is statistically superior to empirical therapy. The benefit of establishing a microbial etiology may be questioned, particularly in light of the cost of diagnostic testing. However, a number of reasons exist for attempting an etiologic diagnosis. Identification of a specific or unexpected pathogen allows narrowing of the initial empirical regimen, with a consequent decrease in antibiotic selection pressure and in the risk of resistance. Pathogens with important public safety implications, such as *Mycobacterium tuberculosis* and influenza virus, may be found. Finally, without susceptibility data, trends in resistance cannot be followed accurately, and appropriate empirical therapeutic regimens are harder to devise.

GRAM'S STAIN AND CULTURE OF SPUTUM The main purpose of the sputum Gram's stain is to ensure suitability of a specimen for culture. (To be suitable, a sputum sample must have >25 neutrophils and <10 squamous epithelial cells per low-power field.) However, staining may also identify certain pathogens (e.g., *S. pneumoniae*, *S. aureus*, and gram-negative bacteria). The sensitivity and specificity of the sputum Gram's stain and culture are highly variable. Even in cases of proven bacteremic pneumococcal pneumonia, the yield of positive cultures from sputum is ≤50%.

Many patients, particularly elderly individuals, may be unable to produce an appropriate sputum sample. Others may be taking antibiotics that interfere with culture results. Inability to produce sputum can be caused by dehydration, whose correction may result in increased

sputum production and a more obvious infiltrate on chest radiography. For patients admitted to the ICU and intubated, a deep-suction aspirate or bronchoalveolar lavage sample has a high yield on culture when sent to the laboratory as soon as possible. Since pathogens in severe and mild CAP may differ (Table 126-1), the greatest benefit of staining and culturing respiratory secretions is to alert the physician to unexpected and/or resistant pathogens and to permit appropriate modification of therapy. Other stains and cultures (e.g., for *M. tuberculosis* or fungi) may be useful as well. The sputum Gram's stain and culture are recommended only for hospitalized CAP patients, particularly those with severe cases or those with risks of MRSA or *P. aeruginosa* infection.

BLOOD CULTURES The yield from blood cultures, even when samples are collected before antibiotic therapy, is disappointingly low. Only 5–14% of cultures from hospitalized CAP patients are positive, and the most common pathogen is *S. pneumoniae*. Since recommended empirical regimens all provide pneumococcal coverage, a blood culture positive for this pathogen has little, if any, effect on clinical outcome. However, susceptibility data may allow narrowing of antibiotic therapy in appropriate cases. Because of the low yield and the lack of significant impact on outcome, blood cultures are not considered *de rigueur* for all hospitalized CAP patients. Certain high-risk patients should have blood cultured, including those with neutropenia secondary to pneumonia, asplenia, complement deficiencies, chronic liver disease, or severe CAP and those at risk of MRSA or *P. aeruginosa* infection.

URINARY ANTIGEN TESTS Two commercially available tests detect pneumococcal and *Legionella* antigen in urine. The *Legionella pneumophila* test detects only serogroup 1, which accounts for most community-acquired cases of Legionnaires' disease in the United States. The sensitivity and specificity of this antigen test are 70% and 99%, respectively. The pneumococcal urine antigen test also is quite sensitive and specific (70% and >90%, respectively). Although false-positive results can be obtained for pneumococcus-colonized children, the test is generally reliable. Both tests can detect antigen even after the initiation of appropriate antibiotic therapy. Testing of urine for pneumococcal antigen can be reserved for severe cases; *Legionella* antigen can be sought in severe cases and in situations where relevant epidemiologic factors are present.

POLYMERASE CHAIN REACTION PCR tests amplify a microorganism's DNA or RNA, and multiplex PCR panels test for a number of viral and bacterial pathogens. These tests dramatically improve response times, but the contamination of respiratory specimens by upper-airway flora may make semiquantitative or quantitative assays necessary for best results. PCR of nasopharyngeal swabs has become the standard for diagnosis of respiratory viral infection. PCR can also detect the nucleic acid of *Legionella* species, *M. pneumoniae*, *C. pneumoniae*, and mycobacteria. The cost-effectiveness of PCR testing, however, has not been definitively established.

SEROLOGY A fourfold rise in specific IgM antibody titer between acute- and convalescent-phase serum samples is generally considered diagnostic of infection with a particular pathogen. Until recently, serologic tests were used to help identify atypical pathogens as well as selected unusual organisms such as *Coxiella burnetii*. However, these tests have fallen out of favor because of the time required to obtain a final result for the convalescent-phase sample and the difficulty of interpretation.

BIOMARKERS Two of the most commonly used markers are C-reactive protein (CRP) and procalcitonin (PCT). Levels of these acute-phase reactants increase in the presence of an inflammatory response, particularly to bacterial pathogens. Nevertheless, PCT is insufficiently accurate for use in the diagnosis of bacterial CAP, and initial serum PCT levels should not be used as a basis for withholding initial antibiotic treatment. CRP is considered even less sensitive than PCT for detecting bacterial pathogens. Thus these tests should not be used alone but, in conjunction with findings from the history, physical examination, radiography, and laboratory tests, may facilitate antibiotic stewardship and appropriate management of seriously ill CAP patients.

TREATMENT

Community-Acquired Pneumonia

SITE OF CARE

The decision to hospitalize a patient with CAP has considerable implications. The cost of inpatient management exceeds that of outpatient treatment by a factor of 20, and hospitalization accounts for most CAP-related expenditures. However, late admission to the ICU is associated with increased mortality rates. The choice can be difficult: some patients can be managed at home, while others require hospitalization. Tools that objectively assess the risk of adverse outcomes, including severe illness and death, can help to minimize unnecessary hospital admissions. The two most frequently used rules are the Pneumonia Severity Index (PSI), a prognostic model that identifies patients at low risk of dying, and the CURB-65 criteria, which yield a severity-of-illness score.

To determine the PSI, points are given for 20 variables, including age, coexisting illness, and abnormal physical and laboratory findings. On the basis of the score, patients are assigned to one of five classes with these mortality rates: class 1, 0.1%; class 2, 0.6%; class 3, 2.8%; class 4, 8.2%; and class 5, 29.2%. Use of the PSI results in lower admission rates for class 1 and class 2 patients. Class 3 patients could ideally be admitted to an observation unit pending further decisions.

The CURB-65 criteria include five variables: confusion (C); urea >7 mmol/L (U); respiratory rate ≥30/min (R); blood pressure—systolic ≤90 mmHg or diastolic ≤60 mmHg (B); and an age of ≥65 years. Patients with a score of 0 (a 30-day mortality rate of 1.5%) can be treated as outpatients. With a score of 1 or 2, the patient should be hospitalized unless the score is entirely or in part attributable to an age of ≥65 years; in such cases, hospitalization may not be necessary. Among patients with scores of ≥3, mortality rates are 22% overall; these patients may require ICU admission. The PSI has greater efficacy than CURB-65 but is more difficult to calculate.

If a patient is unable to maintain oral intake, if compliance is thought to be an issue when assessed on the basis of mental condition or living situation (e.g., cognitive impairment or homelessness), or if the patient's O₂ saturation on room air is <92%, hospitalization is necessary. If these considerations do not apply, clinical judgment in conjunction with a prediction rule should be used to determine the site of care.

Neither PSI nor CURB-65 is accurate in determining the need for ICU admission. Patients with septic shock requiring vasopressors or with acute respiratory failure requiring intubation and mechanical ventilation should be admitted directly to an ICU (Table 126-3), and those with three of the nine minor criteria listed in the latter table should be admitted to an ICU or a high-level monitoring unit. Mortality rates are higher among less ill patients who were admitted

TABLE 126-3 Criteria for Severe Community-Acquired Pneumonia

Minor criteria
Respiratory rate ≥30 breaths/min
PaO ₂ /FiO ₂ ratio ≤250
Multilobar infiltrates
Confusion/disorientation
Uremia (BUN level ≥20 mg/dL)
Leukopenia (WBC count <4000 cells/μL)
Thrombocytopenia (platelet count <100,000 cells/μL)
Hypothermia (core temperature <36°C)
Hypotension requiring aggressive fluid resuscitation
Major criteria
Respiratory failure requiring invasive mechanical ventilation
Septic shock requiring vasopressors

Abbreviations: BUN, blood urea nitrogen; PaO₂/FiO₂, arterial oxygen pressure/fraction of inspired oxygen; WBC, white blood cell.

to a medical floor but then deteriorated than among equally ill patients initially monitored in the ICU.

ANTIBIOTIC RESISTANCE

Antimicrobial resistance is a significant problem that threatens to diminish our therapeutic armamentarium. Antibiotic misuse results in increased antibiotic selection pressure that can affect resistance locally and globally by clonal dissemination. For CAP, the main resistance issues currently involve *S. pneumoniae* and CA-MRSA.

S. pneumoniae In general, pneumococcal resistance to β -lactams is acquired by (1) direct DNA incorporation and remodeling of penicillin-binding proteins through contact with closely related oral commensal bacteria (e.g., viridans group streptococci), (2) the process of natural transformation, or (3) mutation of certain genes.

The *S. pneumoniae* minimal inhibitory concentration (MIC) breakpoint cutoffs for penicillin in pneumonia are ≤ 2 $\mu\text{g/mL}$ for susceptible, >2 – 4 $\mu\text{g/mL}$ for intermediate, and ≥ 8 $\mu\text{g/mL}$ for resistant. A change in susceptibility thresholds dramatically decreased the proportion of pneumococcal isolates considered nonsusceptible. For meningitis, MIC thresholds remain at the former lower levels. Fortunately, resistance to penicillin appeared to plateau even before the change in MIC thresholds. Of isolates in the United States, $<20\%$ are resistant to penicillins and $<1\%$ to cephalosporins. Risk factors for penicillin-resistant pneumococcal infection include recent antimicrobial therapy, an age of <2 or >65 years, attendance at a day-care center, recent hospitalization, and HIV infection.

In contrast to penicillin resistance, macrolide resistance is increasing in *S. pneumoniae* through several mechanisms. *Target-site modification* caused by ribosomal methylation in 23S rRNA encoded by the *ermB* gene results in high-level resistance (MIC, ≥ 64 $\mu\text{g/mL}$) to macrolides, lincosamides, and streptogramin B-type antibiotics. The *efflux mechanism* encoded by the *mef* gene (*M phenotype*) is usually associated with low-level resistance (MIC, 1–32 $\mu\text{g/mL}$). These two mechanisms account for $\sim 40\%$ and $\sim 60\%$, respectively, of resistant pneumococcal isolates in the United States. High-level resistance to macrolides is more common in Europe, whereas lower-level resistance predominates in North America. The prevalence of macrolide-resistant *S. pneumoniae* exceeds 25% in some countries; in Canada the prevalence is $\sim 22\%$, and in the United States it exceeds 30%. Much of this resistance is high-level, and failures of treatment may result in such cases. In these situations, a macrolide should not be used as empirical monotherapy. Estimates of the prevalence of doxycycline resistance in the United States are generally $<20\%$.

The rate of pneumococcal resistance to fluoroquinolones (e.g., ciprofloxacin, moxifloxacin, and levofloxacin) is usually $<2\%$. Changes can occur in one or both target sites (topoisomerases II and IV); these changes are attributable to mutations in the *gyrA* and *parC* genes, respectively. In addition, an efflux pump may play a role in pneumococcal resistance to fluoroquinolones.

Isolates resistant to drugs from three or more antimicrobial classes with different mechanisms of action are considered MDR strains. The propensity for an association of pneumococcal resistance to penicillin with reduced susceptibility to other drugs, such as macrolides, tetracyclines, and trimethoprim-sulfamethoxazole, is of concern. In the United States, 58.9% of penicillin-resistant pneumococcal blood isolates are also resistant to macrolides.

The most important risk factor for antibiotic-resistant pneumococcal infection is use of a specific antibiotic within the previous 3 months. A history of prior antibiotic treatment is a critical factor in avoiding the use of an inappropriate antibiotic.

CA-MRSA CAP due to MRSA may be caused by the classic hospital-acquired strains or by genotypically and phenotypically distinct community-acquired strains. Most infections with the former have been acquired either directly or indirectly during contact with the health care environment. However, in some hospitals, CA-MRSA strains are displacing the classic hospital-acquired

strains; this change suggests that the newer community-acquired strains may be more robust.

Methicillin resistance in *S. aureus* is determined by the *mecA* gene, which encodes for resistance to all β -lactam drugs. At least five staphylococcal chromosomal cassette *mec* (SCC*mec*) types have been described. The typical hospital-acquired strain usually has a type II or III SCC*mec* element, whereas CA-MRSA has type IV. CA-MRSA isolates tend to be less resistant than the older hospital-acquired strains and are often susceptible to trimethoprim-sulfamethoxazole, clindamycin, and tetracycline in addition to vancomycin and linezolid. However, the most important distinction is that CA-MRSA strains also carry genes for superantigens such as enterotoxins B and C and Panton-Valentine leukocidin; the latter is a membrane-tropic toxin that can create cytolytic pores in neutrophils, monocytes, and macrophages.

M. pneumoniae Macrolide-resistant *M. pneumoniae* has been reported in a number of countries, including Germany (3%), Japan (30%), China (95%), and France and the United States (5–13%). *Mycoplasma* resistance to macrolides is increasing as a result of binding-site mutation in domain V of 23S rRNA.

Gram-Negative Bacilli A detailed discussion of resistance among gram-negative bacilli is beyond the scope of this chapter (see Chap. 161). Fluoroquinolone resistance among community isolates of *Escherichia coli* is increasing. *Enterobacter* species are typically resistant to cephalosporins, and the drugs of choice for use against these organisms are usually fluoroquinolones or carbapenems. Similarly, when infections due to bacteria producing extended-spectrum β -lactamases (ESBLs) are documented or suspected, a carbapenem should be considered.

INITIAL ANTIBIOTIC MANAGEMENT

Since the etiology of CAP is rarely known at the outset of treatment, initial therapy is usually empirical and designed to cover the likeliest pathogens. In all cases, treatment should be initiated as expeditiously as possible. New CAP treatment guidelines in the United States have been presented in a joint statement from the American Thoracic Society (ATS) and the Infectious Diseases Society of America (IDSA). These guidelines consider the likely pathogens, risk of antimicrobial resistance, severity of illness, site of care, and risk of infection with specific bacteria such as MRSA and *P. aeruginosa* (Fig. 126-1, Tables 126-4 and 126-5). In the figure and the tables, the antibiotics are not listed in order of preference.

The approach to treatment of aspiration pneumonia is based upon a number of factors, including site of acquisition (community

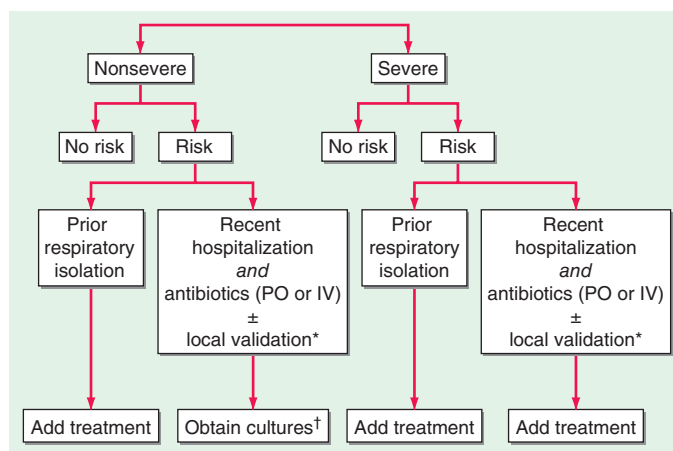


FIGURE 126-1 Algorithm for assessment of inpatient risk of infection with MRSA or *Pseudomonas aeruginosa*. Underlying lung disease (e.g., bronchiectasis or very severe COPD) are also risks for *P. aeruginosa* infection. *Local validation consists of information on local prevalence, resistance, and risk factors. †Can also use MRSA rapid nasal PCR if available.

TABLE 126-4 Initial Treatment Strategies for Outpatients with Community-Acquired Pneumonia

STATUS	STANDARD REGIMEN
No comorbidities or risk factors for antibiotic resistance ^a	Combination therapy with amoxicillin (1 g tid) + either a macrolide ^b or doxycycline (100 mg bid) <i>Or</i> Monotherapy with doxycycline (100 mg bid) <i>Or</i> Monotherapy with a macrolide ^{b,c}
With comorbidities ^d ± risk factors for antibiotic resistance ^a	Combination therapy with amoxicillin/clavulanate ^e or a cephalosporin ^f + either a macrolide ^b or doxycycline (100 mg bid) <i>Or</i> Monotherapy with a respiratory fluoroquinolone ^g

^aAntibiotic treatment within the past 3 months or contact with the health care system. ^bAzithromycin (500 mg on day 1, then 250 mg/d for 4 days), clarithromycin (500 mg bid), or clarithromycin ER (1000 mg/d). ^cIf local prevalence of pneumococcal resistance is <25%. ^dIncluding chronic heart, lung, liver, or kidney disease; diabetes mellitus; alcoholism; malignancy; or asplenia. ^e500/125 mg tid or 875/125 mg bid. ^fCefpodoxime (200 mg bid) or cefuroxime (500 mg bid). ^gLevofloxacin (750 mg/d), moxifloxacin (400 mg/d), or gemifloxacin (320 mg/d).

vs hospital), normal or abnormal chest radiograph, and additional variables such as illness severity, state of dentition, and risk of infection with an MDR pathogen. Routine coverage of anaerobes is unnecessary unless dentition is poor or there is a lung abscess or necrotizing pneumonia.

Our approach to the treatment of CAP (Tables 126-4 and 126-5) is very similar to that proposed in the new CAP guidelines with the exceptions listed below.

Outpatients The exceptions to the CAP guidelines that we follow in treating patients are:

- We usually initiate coverage that includes atypical organisms as well as *S. pneumoniae*.
- Generally, we do not consider the risk of infection with *P. aeruginosa* or MRSA particularly significant in outpatients.
- Prior antibiotic use should include both oral and parenteral agents.

TABLE 126-5 Initial Treatment for Inpatients with or without Risk Factors for Infection with MRSA or *Pseudomonas aeruginosa*

DISEASE SEVERITY, RISK STATUS	REGIMEN
Nonsevere	
No risk factors	A β-lactam ^a + a macrolide ^b <i>or</i> A respiratory fluoroquinolone ^c
Prior respiratory isolation	Add coverage for MRSA ^d or <i>Pseudomonas aeruginosa</i> ^e
Recent hospitalization, antibiotic treatment, ± LV ^f	Add coverage for MRSA ^d or <i>P. aeruginosa</i> ^e only if cultures are positive
Severe	
No risk factors	A β-lactam ^a + a macrolide ^b <i>or</i> A β-lactam ^a + respiratory fluoroquinolone ^c
Prior respiratory isolation	Add coverage for MRSA ^d or <i>P. aeruginosa</i> ^e
Recent hospitalization, antibiotic treatment ± LV ^f	Add coverage for MRSA ^d or <i>P. aeruginosa</i> ^e

^aAmpicillin-sulbactam (1.5–3 g q6h). ^bAzithromycin (500 mg/d) or clarithromycin (500 mg bid). ^cLevofloxacin (750 mg/d), moxifloxacin (400 mg/d), or gemifloxacin (320 mg/d). ^dVancomycin (15 mg/kg q12h, with adjustment based on serum levels) or linezolid (600 mg q12h). ^ePiperacillin-tazobactam (4.5 g q6h), cefepime (2 g q8h), ceftazidime (2 g q8h), imipenem (500 mg q6h), meropenem (1 g q8h), or aztreonam (2 g q8h). ^fObtain cultures. MRSA rapid nasal PCR can also be used if available.

Abbreviations: LV, local validation (local prevalence, resistance, risk factors); MRSA, methicillin-resistant *Staphylococcus aureus*.

Patients are stratified into two groups: those without comorbidity or risk factors for antibiotic resistance and those with comorbidities (e.g., chronic heart, lung, liver, or kidney disease; diabetes; alcoholism; malignancy; or asplenia) with or without risk factors for resistance (Table 126-4). As a general rule, if patients have been treated with a drug from a particular class of antibiotics within the previous 3 months, drugs from a different class should be used to minimize resistance issues.

For those without comorbidity or resistance risk factors, amoxicillin alone or doxycycline is recommended in the recent guidelines. Monotherapy with amoxicillin is based on evidence of its efficacy in the treatment of hospitalized CAP patients. This recommendation is a change from that in the 2007 IDSA/ATS CAP guidelines. As a rule, however, we usually tend to initiate treatment that includes coverage for *S. pneumoniae* as well as the atypical pathogens (Table 126-4).

Monotherapy with a macrolide is recommended in the new guidelines only if there are contraindications to amoxicillin or doxycycline and there is documented low risk of macrolide resistance (<25%). Otherwise, the treatment of outpatients is quite similar to the regimens recommended in the 2007 IDSA/ATS guidelines.

Inpatients Our exceptions to the recommendations in the CAP guidelines are:

- As a general rule, when initiating treatment for infection with *P. aeruginosa*, we use double coverage.
- The presence of all three risk factors is not required for drug resistance (recent hospitalization, recent oral or IV antibiotic treatment, ± local validation) (Fig. 126-1, Table 126-5).

The main considerations for determining initial empirical treatment of hospitalized CAP patients are clinical severity and risk factors for infection with drug-resistant pathogens such as MRSA or *P. aeruginosa*. Hospitalization alone is not now considered a significant risk factor for these pathogens. Hospitals should collect local data on MRSA and *P. aeruginosa* with regard to prevalence, risk factors for infection, and antibiotic susceptibilities. Patients can be categorized as having nonsevere or severe CAP (Table 126-3), and those in each of these categories may or may not have risk factors for MRSA or *P. aeruginosa* (Fig. 126-1). In the scenarios involving these variables in hospitalized CAP patients, empirical treatment for either of these pathogens should be added to standard therapy unless a patient's illness is considered nonsevere and the risk factors are recent hospitalization and antibiotic treatment ± local validation data (Fig. 126-1). Depending upon the patient, we may begin treatment in this situation and then de-escalate it if appropriate. In such cases, cultures should be performed but treatment usually withheld unless the culture results or the rapid nasal PCR results for MRSA are positive.

Nonsevere, No Risk Factors For patients with nonsevere infection and no risk factors, treatment should consist of either a combination of a β-lactam and a macrolide or monotherapy with a respiratory fluoroquinolone (Table 126-5). In the event of contraindications to macrolides and fluoroquinolones, a β-lactam together with doxycycline may be used. Treatment with a combination of a β-lactam and a macrolide or a fluoroquinolone alone results in lower mortality than monotherapy with a β-lactam.

Severe, No Risk Factors Patients with severe infection but no risk factors should receive combination therapy with either a β-lactam and a macrolide or a β-lactam and a respiratory fluoroquinolone (Table 126-5).

Nonsevere and Severe, with Risk Factors To date, there are no prediction rules reliably identifying patients who should be started empirically on treatment for MRSA or *P. aeruginosa*. Current risk factors for infection with these pathogens are hierarchical. Prior isolation of these organisms, especially from the respiratory tract within the previous year, is a more robust risk factor than recent hospitalization and exposure to parenteral antibiotics. For

P. aeruginosa, underlying lung disease (e.g., bronchiectasis or very severe COPD) also is an important risk factor. If MRSA or *P. aeruginosa* has been isolated previously, appropriate empirical therapy should be started in both severe and nonsevere cases (Table 126-5). We prefer linezolid over vancomycin as first-line treatment for MRSA because of its inhibition of bacterial exotoxin and its better lung penetration. If the organism is not isolated from respiratory secretions or blood and/or the nasal or bronchoalveolar lavage PCR test for MRSA is negative and the patient is improving at 48 h, treatment may be de-escalated to a standard regimen.

If, on the other hand, the risk factors are recent hospitalization and antibiotic use within the previous 3 months, appropriate samples should be obtained for culture, and, in severe cases, extended-spectrum treatment for MRSA or *P. aeruginosa* should be initiated. Depending upon the severity of infection, local data on *P. aeruginosa* resistance, and antibiotic use within the previous 90 days, single- or double-drug coverage should be used.

If two antipseudomonal agents are started, the drugs should not be from the same class. Whenever possible, assessment for possible de-escalation of therapy is urged. If the patient's illness is not severe, empirical extended treatment should be withheld until culture results are available.

Regardless of the site of care, CAP patients testing positive for influenza should be given anti-influenza treatment (e.g., oseltamivir) as well as appropriate antibacterial therapy. Physicians should be vigilant about possible superinfection with MRSA.

Although hospitalized patients have traditionally received initial therapy by the IV route, some drugs, particularly the fluoroquinolones, are very well absorbed and may be given orally from the outset to select patients. For those initially treated with IV agents, a switch to oral treatment is appropriate when the patient can ingest and absorb the drugs, is hemodynamically stable, and is showing clinical improvement. A 5-day course of treatment is usually sufficient for uncomplicated CAP, but longer treatment may be required for patients who have not stabilized clinically and for those with bacteremia, metastatic infection, or infection with a more virulent pathogen such as *P. aeruginosa* or MRSA.

ADJUNCTIVE MEASURES

In addition to appropriate antimicrobial therapy, certain adjunctive measures should be used. Adequate hydration, oxygen therapy for hypoxemia, vasopressor treatment, and assisted ventilation when necessary are critical to successful treatment. Routine use of glucocorticoids is not recommended for CAP except in patients with refractory septic shock.

FAILURE TO IMPROVE

Patients slow to respond to therapy should be reevaluated at about day 3 (sooner if their condition is worsening), with several scenarios considered. A number of noninfectious conditions mimic pneumonia, including pulmonary edema, pulmonary embolism, lung carcinoma, radiation and hypersensitivity pneumonitis, and connective tissue disease involving the lungs. If the patient truly has CAP and empirical treatment is aimed at the correct pathogen, lack of response may be explained in a number of ways. The pathogen may be resistant to the drug selected, or a sequestered focus (e.g., lung abscess or empyema) may prevent antibiotic access to the pathogen. The patient may be getting the wrong drug or the correct drug at the wrong dose or frequency of administration. Another possibility is that CAP has been diagnosed correctly but an unexpected pathogen (e.g., CA-MRSA, *M. tuberculosis*, or a fungus) is the cause. Nosocomial superinfections—both pulmonary and extrapulmonary—are other possible explanations for a hospitalized patient's failure to improve. In all cases of delayed response or worsening condition, the patient must be carefully reassessed and appropriate studies initiated, possibly including CT or bronchoscopy.

COMPLICATIONS

Complications of severe CAP include respiratory failure, shock and multiorgan failure, and exacerbation of comorbid illnesses. Three

particularly noteworthy conditions are metastatic infection, lung abscess, and complicated pleural effusion. Metastatic infection (e.g., brain abscess or endocarditis) is unusual and requires a high degree of suspicion and a detailed workup for proper treatment. Lung abscess may occur in association with aspiration pneumonia or with infection caused by pathogens such as CA-MRSA, *P. aeruginosa*, or (rarely) *S. pneumoniae*. A significant pleural effusion should be tapped for both diagnostic and therapeutic purposes. If the fluid has a pH <7.2, a glucose level of <2.2 mmol/L, and a lactate dehydrogenase concentration of >1000 U/L or if bacteria are seen or cultured, drainage is needed.

FOLLOW-UP

Fever and leukocytosis usually resolve within 2–4 days in otherwise healthy patients with CAP, but physical findings may persist longer. Chest radiographic abnormalities are slowest to resolve (4–12 weeks), with the speed of clearance depending on the patient's age and underlying lung disease. Patients may be discharged from the hospital once their clinical condition, including any comorbidity, is stable. The site of residence after discharge (nursing home, home with family, home alone) is an important consideration, particularly for elderly patients. For a hospitalized patient, we generally recommend a follow-up radiograph ~4–6 weeks later. If relapse or recurrence is documented, particularly in the same lung segment, the possibility of an underlying neoplasm must be considered. For individuals managed as outpatients, routine follow-up chest radiography is not necessary if they are nonsmokers, if they are otherwise well, and if their symptoms resolved within 5–7 days.

PROGNOSIS

The prognosis depends on the patient's age, comorbidities, and site of treatment (inpatient or outpatient). Young patients without comorbidity do well and usually recover fully after ~2 weeks. Older patients and those with comorbid conditions may take several weeks longer to recover fully. The overall mortality rate for the outpatient group is <5%. For patients requiring hospitalization, overall mortality ranges from 12% to 40%, depending on the category of patient and the processes of care, particularly the timely administration of appropriate antibiotics.

PREVENTION

The main preventive measure is vaccination (**Chap. 123**). Recommendations of the Advisory Committee on Immunization Practices should be followed for influenza and pneumococcal vaccines.

A pneumococcal polysaccharide vaccine (PPSV23) and a protein conjugate pneumococcal vaccine (PCV13) are available in the United States (**Chap. 146**). The former contains capsular material from 23 pneumococcal serotypes; in the latter, capsular polysaccharide from 13 of the most common pneumococcal pathogens affecting children is linked to an immunogenic protein. PCV13 produces T-cell-dependent antigens, resulting in long-term immunologic memory. Administration of this vaccine to children has led to a decrease in the prevalence of antimicrobial-resistant pneumococci and in the incidence of invasive pneumococcal disease among both children and adults. However, vaccination can result in the replacement of vaccine with nonvaccine serotypes, as seen with serotypes 19A and 35B following introduction of the original 7-valent conjugate vaccine. PCV13 is also recommended for the elderly and for younger immunocompromised patients. Because of an increased risk of pneumococcal infection, even among patients without obstructive lung disease, smokers should be strongly encouraged to quit.

The influenza vaccine is available in an inactivated or recombinant form. During an influenza outbreak, unprotected patients at risk from complications should be vaccinated immediately and given chemoprophylaxis with either oseltamivir or zanamivir for 2 weeks—i.e., until vaccine-induced antibody levels are sufficiently high.

VENTILATOR-ASSOCIATED PNEUMONIA

Research on hospital-acquired pneumonia has focused on VAP. However, the same information and principles can also be applied to ventilated HAP and to non-ICU HAP. Approximately 70% of HAP cases are

1016 acquired outside the ICU and 30% in the ICU; the fact that 30% of all HAP patients need mechanical ventilation defines ventilated HAP as a distinct entity. In non-intubated patients with HAP, an expectorated sputum sample is used for microbiologic diagnosis, but results are confounded by frequent colonization by oral pathogens. Microbiologic information in VAP and ventilated HAP is obtained from direct access to deep lower respiratory tract samples, which provide reliable microbiologic data; however, these samples can also contain colonizing pathogens.

ETIOLOGY

Potential etiologic agents of VAP include both MDR and non-MDR bacterial pathogens (Table 126-6). The non-MDR group of “core pathogens” is nearly identical to the pathogens found in severe CAP (Table 126-1); it is not surprising that such pathogens predominate if VAP develops in the first 5–7 days of the hospital stay. However, if patients have other risk factors (particularly prior antibiotic treatment), MDR pathogens are a consideration, even early in the hospital course. The relative frequency of individual MDR pathogens can vary significantly from hospital to hospital and even between different critical care units within the same institution. Most hospitals have problems with *P. aeruginosa* and MRSA, but other MDR pathogens are often institution-specific. Less commonly, fungal and viral pathogens cause VAP, usually affecting severely immunocompromised patients. Rarely, community-associated viruses cause mini-epidemics, usually when introduced by ill health care workers.

EPIDEMIOLOGY

Pneumonia is a common complication among patients requiring mechanical ventilation. Prevalence estimates vary between 6 and 52 cases per 100 patients, depending on the population studied. On any given day in the ICU, an average of 10% of patients will have pneumonia—VAP in the overwhelming majority of cases, although in recent years the frequency of this infection is declining as a result of effective prevention strategies. The frequency of diagnosis is not static but changes with the duration of mechanical ventilation, with the highest hazard ratio in the first 5 days and a plateau in additional cases (1% per day) after ~2 weeks. However, the cumulative rate among patients who remain ventilated for as long as 30 days is as high as 70%. These rates often do not reflect the recurrence of VAP in the same patient. Once a ventilated patient is transferred to a chronic-care facility or to home, the incidence of pneumonia drops significantly, especially in the absence of other risk factors for pneumonia. However, in chronic ventilator units, purulent tracheobronchitis becomes a significant issue, often interfering with efforts to wean patients off mechanical ventilation (Chap. 302).

Three factors are critical in the pathogenesis of VAP: colonization of the oropharynx with pathogenic microorganisms, aspiration of these organisms from the oropharynx into the lower respiratory tract, and compromise of normal host defense mechanisms. Most risk factors and

TABLE 126-7 Pathogenic Mechanisms and Corresponding Prevention Strategies for Ventilator-Associated Pneumonia

PATHOGENIC MECHANISM	PREVENTION STRATEGY
Oropharyngeal colonization with pathogenic bacteria	
Elimination of normal flora, overgrowth by pathogenic bacteria	Avoidance of prolonged antibiotic courses; consider oral chlorhexidine ^a
Large-volume oropharyngeal aspiration around time of intubation	Short course of prophylactic antibiotics for comatose patients ^b
Gastroesophageal reflux	Postpyloric enteral feeding with orally placed feeding tube ^c ; avoidance of high gastric residuals, prokinetic agents
Bacterial overgrowth of stomach	Avoidance of prophylactic agents that raise gastric pH ^d ; selective decontamination of digestive tract with nonabsorbable antibiotics ^a
Cross-infection from other colonized patients	Hand washing, especially with alcohol-based hand rub; intensive infection control education ^b ; isolation; proper cleaning of reusable equipment
Large-volume aspiration	Endotracheal intubation; rapid-sequence intubation technique; avoidance of sedation; decompression of small-bowel obstruction
Ventilator circuit humidification	Change ventilator circuits only when soiled and with new patient; drain ventilator circuit condensate away from patient; replace heat moisture exchanger every 5–7 days or if soiled or malfunctioning ^a
Microaspiration around endotracheal tube	
Endotracheal intubation	Noninvasive ventilation ^b
Prolonged duration of ventilation	Daily awakening from sedation, ^b weaning protocols ^b
Abnormal swallowing function	Early percutaneous tracheostomy ^b
Secretions pooled above endotracheal tube	Head of bed elevated ^b ; continuous aspiration of subglottic secretions with specialized endotracheal tube ^b ; avoidance of reintubation; minimization of sedation and patient transport; prophylactic PEEP ^c of 5–8 cm
Altered lower respiratory host defenses	Tight glycemic control ^b ; lowering of hemoglobin transfusion threshold

^aStrategies with negative randomized trials or conflicting results. ^bStrategies demonstrated to be effective in at least one randomized controlled trial. ^cPositive end-expiratory pressure.

their corresponding prevention strategies pertain to one of these three factors (Table 126-7).

The most obvious risk factor is the endotracheal tube, which bypasses the normal mechanical factors preventing aspiration. While the presence of an endotracheal tube may prevent large-volume aspiration, microaspiration is actually exacerbated by secretions pooling above the cuff. The endotracheal tube and the concomitant need for suctioning can damage the tracheal mucosa, thereby facilitating tracheal colonization. In addition, pathogenic bacteria can form a glycocalyx biofilm on the tube's surface that protects them from both antibiotics and host defenses. The bacteria can also be dislodged during suctioning (done preferably with a closed catheter system) and can re-inoculate the trachea, or tiny fragments of a glycocalyx can embolize to distal airways, carrying bacteria with them. The ventilator circuit tubing can harbor pathogenic organisms that can wash back to the patient if manipulated too often; thus circuits are changed only when soiled and with each new patient. Heat moisture exchangers are changed every 5–7 days or if visibly soiled or malfunctioning.

In a high percentage of critically ill patients, the normal oropharyngeal flora is replaced by pathogenic microorganisms. The most important risk factors are antibiotic selection pressure, cross-infection from other infected/colonized patients or contaminated equipment, severe systemic

TABLE 126-6 Microbiologic Causes of Ventilator-Associated Pneumonia

NON-MDR PATHOGENS	MDR PATHOGENS
<i>Streptococcus pneumoniae</i>	<i>Pseudomonas aeruginosa</i>
Other <i>Streptococcus</i> spp.	Methicillin-resistant <i>S. aureus</i>
<i>Haemophilus influenzae</i>	<i>Acinetobacter</i> spp.
Methicillin-sensitive <i>Staphylococcus aureus</i>	Antibiotic-resistant Enterobacteriaceae
Antibiotic-sensitive Enterobacteriaceae	ESBL-positive strains
<i>Escherichia coli</i>	Carbapenem-resistant strains
<i>Klebsiella pneumoniae</i>	<i>Legionella pneumophila</i>
<i>Proteus</i> spp.	<i>Burkholderia cepacia</i>
<i>Enterobacter</i> spp.	<i>Aspergillus</i> spp.
<i>Serratia marcescens</i>	

Abbreviations: ESBL, extended-spectrum β-lactamase; MDR, multidrug-resistant.

illness, and malnutrition. Of these factors, antibiotic exposure poses the greatest risk by far. Pathogens such as *P. aeruginosa* almost never cause infection in patients without prior exposure to antibiotics. The recent emphasis on hand hygiene has lowered the cross-infection rate.

Almost all intubated patients experience microaspiration and are at least transiently colonized with pathogenic bacteria. However, only around one-third of colonized patients develop VAP. Colony counts increase to high levels, sometimes days before the development of clinical pneumonia; these increases suggest that the final step in VAP development, independent of aspiration and oropharyngeal colonization, is the overwhelming of host defenses by a large bacterial inoculum. Severely ill patients with sepsis and trauma appear to enter a state of immunoparalysis several days after admission to the ICU—a time that corresponds to the greatest risk of developing VAP. The mechanism of this immunosuppression is not clear, although hyperglycemia and frequent transfusions adversely affect the immune response.

■ CLINICAL MANIFESTATIONS

The clinical manifestations of HAP and VAP are nonspecific: fever, leukocytosis, increased respiratory secretions, and pulmonary consolidation on physical examination, along with a new or changing radiographic infiltrate. The frequency of abnormal chest radiographs before the onset of pneumonia in intubated patients and the limitations of portable radiographic technique make interpretation of radiographs more difficult than in patients who are not intubated. Other clinical features may include tachypnea, tachycardia, worsening oxygenation, and increased minute ventilation. Serial changes in oxygenation may identify pneumonia earlier than other findings and may also be a means to monitor improvement with therapy.

■ DIAGNOSIS

No single set of criteria is reliably diagnostic of pneumonia in a ventilated patient. The inability to accurately identify such patients compromises efforts to prevent and treat VAP and even calls into question estimates of the impact of VAP on mortality rates.

Application of the clinical criteria typical for CAP consistently results in overdiagnosis of VAP, largely because of (1) frequent tracheal colonization with pathogenic bacteria in patients with endotracheal tubes, (2) multiple alternative causes of radiographic infiltrates in mechanically ventilated patients, and (3) the high frequency of other sources of fever in critically ill patients. The differential diagnosis of VAP includes atypical pulmonary edema, pulmonary contusion, alveolar hemorrhage, hypersensitivity pneumonitis, acute respiratory distress syndrome, and pulmonary infarction. Findings of fever and/or leukocytosis may have alternative causes, including antibiotic-associated diarrhea, central line-associated infection, sinusitis, urinary tract infection, pancreatitis, and drug fever. Conditions mimicking pneumonia are often documented in patients in whom VAP has been ruled out by accurate diagnostic techniques. Most of these alternative diagnoses do not require antibiotic treatment; require antibiotics different from those used to treat VAP (fungal or viral pneumonia); or require some additional intervention, such as surgical drainage or catheter removal, for optimal management.

This diagnostic dilemma has led to debate and controversy about whether a quantitative-culture approach as a means of eliminating false-positive clinical diagnoses is superior to a clinical approach enhanced by principles learned from quantitative-culture studies. The most recent IDSA/ATS guidelines for HAP/VAP give a weak recommendation for a clinical approach based on semiquantitative cultures, with consideration of the availability of resources, cost, and the availability of expertise. The guidelines acknowledge that the use of a quantitative approach may result in less antibiotic use, which may be critical for antibiotic stewardship in the ICU. Therefore, the approach at each institution—or potentially for each patient—should be individualized and based on local colonization rates, local diagnostic expertise, and recent history of antibiotic therapy.

Quantitative-Culture Approach This method uses quantitative cultures of deep respiratory tract samples to distinguish colonization from true infection. The more distal in the respiratory tree the diagnostic sampling, the more specific the results and therefore the lower

the threshold of growth necessary to diagnose pneumonia and exclude colonization. For example, a quantitative endotracheal aspirate yields proximal samples, and the diagnostic threshold is 10^6 cfu/mL. The protected specimen brush method, in contrast, collects distal samples and has a threshold of 10^3 cfu/mL. Conversely, sensitivity declines as more distal secretions are obtained, especially when they are collected blindly (i.e., by a technique other than bronchoscopy). Additional tests that may increase the diagnostic yield include Gram's staining, differential cell counts, staining for intracellular organisms, and detection of local protein levels elevated in response to infection.

If the quantitative approach is used, therapy decisions should be linked to culture results, with antibiotics withheld until results are available unless the patient is critically ill. Studies have documented less antibiotic use with this approach than with the clinical approach, but the results are less clear if antibiotic decisions are not directly linked to culture data. One common limitation of the quantitative approach is the use of a new and effective antibiotic agent in the 24–48 h prior to sampling, which can lead to false-negative results. With sensitive microorganisms, a single antibiotic dose can reduce colony counts below the diagnostic threshold. After 3 days, the operating characteristics of the tests improve to the point at which they are equivalent to results obtained when no prior antibiotic therapy has been given. Conversely, colony counts above the diagnostic threshold during antibiotic therapy suggest that the current antibiotics are ineffective. In addition, quantitative cultures may give results below the diagnostic threshold if samples are collected early in the course of infection or if sampling is delayed until after an effective host response has reduced bacterial counts. Ideally, a specimen should be obtained as soon as pneumonia is suspected and before antibiotic therapy is initiated or changed.

Clinical Approach The lack of specificity of a clinical diagnosis of VAP has hampered its utility, but this approach has been improved by the addition of microbiologic and other laboratory data. Tracheal aspirates generally yield at least twice as many potential pathogens as quantitative cultures, but the causative pathogen is almost always present. The absence of bacteria in Gram-stained endotracheal aspirates makes pneumonia an unlikely cause of fever or pulmonary infiltrates. These findings, coupled with a heightened awareness of the alternative diagnoses possible in patients with suspected VAP, can prevent inappropriate antibiotic overtreatment. Furthermore, the absence of an MDR pathogen in tracheal aspirate cultures eliminates the need for MDR coverage, allowing de-escalation of empirical antibiotic therapy. Similarly, with newer and more sensitive molecular diagnostic methods, a suspected MDR pathogen can be eliminated as a therapy target if test results are negative. A clinical approach that focuses on careful antimicrobial use and de-escalation of therapy after culture results become available may have an impact on the avoidance of antimicrobial overuse and the consideration of alternative sites of infection similar to that of a quantitative-culture approach.

TREATMENT

Ventilator-Associated Pneumonia

Many studies have demonstrated higher mortality rates with the delay of initially appropriate empirical antibiotic therapy. The key to appropriate antibiotic management of VAP is an appreciation of the resistance patterns of the most likely pathogens in a given patient.

ANTIBIOTIC RESISTANCE

Because of a higher risk of infection with MDR pathogens (Table 126-6), VAP is treated with antibiotics different from those used for severe CAP. Antibiotic selection pressure leads to the frequent involvement of MDR pathogens by selecting either for drug-resistant isolates of common pathogens (MRSA and Enterobacteriaceae producing ESBLs or carbapenemases) or for intrinsically resistant pathogens (*P. aeruginosa* and *Acinetobacter* species). Frequent use of β -lactam drugs, especially cephalosporins, appears to be the major risk factor for infection with MRSA and ESBL-positive strains.

P. aeruginosa can develop resistance to all routinely used antibiotics, and, even if initially sensitive, *P. aeruginosa* isolates may develop resistance during treatment. Either derepression of resistance genes or selection of resistant clones within the large bacterial inoculum associated with most pneumonias may be the cause. *Acinetobacter* species, *Stenotrophomonas maltophilia*, and *Burkholderia cepacia* are intrinsically resistant to many of the empirical antibiotic regimens employed (see below). VAP caused by these pathogens emerges during treatment of other infections, and resistance is always evident at initial diagnosis.

EMPIRICAL THERAPY

Recommended options for empirical therapy are listed in **Table 126-8**. Treatment should be started once diagnostic specimens have been obtained. The major factors in the selection of agents are the presence of risk factors for MDR pathogens and the predicted risk of death ($\leq 15\%$ is considered low risk). Choices among the various options listed depend on local patterns of resistance and—a very important factor—the patient's prior antibiotic exposure. Knowledge of the local hospital's—and even the specific ICU's—antibiogram and the local incidence of specific MDR pathogens (e.g., MRSA) is critical in selecting appropriate empirical therapy.

The majority of patients *without* risk factors for MDR infection can be treated with a single agent. In fact, mortality is lower with a single agent than with combination therapy for those with a low mortality risk. Unfortunately, the proportion of patients with no MDR risk factors is $<10\%$ in some ICUs and is unknown for HAP patients. The major difference from CAP is the markedly lower incidence of atypical pathogens in VAP; the exception is *Legionella*, which can be a nosocomial pathogen, especially with local epidemics due to breakdowns in the treatment of potable water in the hospital. The standard recommendation for patients *with* risk factors for MDR infection and a high mortality risk is for three antibiotics: two directed at *P. aeruginosa* and one at MRSA. However, in the absence of septic shock, a single agent may be effective for these patients, provided there is a single agent that is likely to be effective against at least 90% of the gram-negative pathogens in that ICU. Empirical combination therapy enhances the likelihood of initially appropriate therapy over that with monotherapy. A β -lactam agent provides the greatest coverage, yet even the broadest-

spectrum agent—a carbapenem—still constitutes inappropriate initial therapy in up to 10–15% of cases at some centers. The emergence of carbapenem resistance at some institutions requires the addition of polymyxins to the combination-therapy options. A number of emerging agents may modify our approach to therapy. New antipseudomonal agents include ceftazidime–avibactam, ceftolozane–tazobactam, imipenem–relebactam, and plazomicin. Therapy for carbapenem-resistant Enterobacteriaceae can consist of ceftazidime–avibactam, imipenem–relebactam, or meropenem–vaborbactam, while organisms that produce metallo- β -lactamases can be treated with ceftazidime–avibactam or cefiderocol.

SPECIFIC TREATMENT

Once an etiologic diagnosis is made, broad-spectrum empirical therapy can be modified (de-escalated) to specifically address the known pathogen. For patients with MDR risk factors, antibiotic regimens can be reduced to a single agent in most cases. Only a minority of cases require a complete course with two or three drugs. A negative tracheal-aspirate culture or growth below the threshold for quantitative cultures of samples obtained before any antibiotic change strongly suggests that antibiotics should be discontinued or that an alternative diagnosis should be pursued. Identification of other confirmed or suspected sites of infection may require ongoing antibiotic therapy, but the spectrum of pathogens (and the corresponding antibiotic choices) may be different from those for VAP. A 7- or 8-day course of therapy is just as effective as a 2-week course and is associated with less frequent emergence of antibiotic-resistant strains. Exceptions include cases in which initial therapy is inappropriate or consists of second-line antibiotics and cases caused by some more resistant organisms, such as carbapenemase-producing *Acinetobacter* species.

A major controversy regarding specific therapy for VAP concerns the need for ongoing combination treatment of *Pseudomonas* pneumonia. No randomized controlled trials have demonstrated a benefit of combination therapy with a β -lactam and an aminoglycoside, nor have subgroup analyses in other trials found a survival benefit with such a regimen. Combination therapy may have value in bacteremic infection with septic shock, but the benefit may last for only a few days. The unacceptably high rates of clinical failure and death despite combination therapy among patients with VAP caused by *P. aeruginosa* (see “Failure to Improve,” below) indicate that better regimens are needed, perhaps including aerosolized antibiotics. In most cases of *Pseudomonas* pneumonia, current guidelines recommend against continuing combination therapy after the isolate's microbial susceptibility is known.

FAILURE TO IMPROVE

Treatment failure is not uncommon in VAP, especially that caused by MDR pathogens. VAP caused by MRSA is associated with a 40% clinical failure rate when treated with standard-dose vancomycin. One proposed but unproven solution is the use of high-dose individualized treatment, although the risk of renal toxicity increases with this strategy. In addition, the MIC of MRSA to vancomycin has been increasing, and a high percentage of clinical failures occur when the MIC is in the upper range of sensitivity (i.e., 1.5–2 $\mu\text{g/mL}$). Linezolid appears to be 15% more efficacious than even adjusted-dose vancomycin and is preferred in patients with renal insufficiency and those infected with high-MIC isolates of MRSA. VAP due to *Pseudomonas* has a 40–50% failure rate, no matter what the regimen. Causes of clinical failure vary with the pathogen(s) and the antibiotic(s). Inappropriate initial therapy can usually be minimized by use of the recommended combination regimen (Table 126-8). However, the emergence of β -lactam resistance during therapy is an important problem, especially in infection with *Pseudomonas* and *Enterobacter* species. Recurrent VAP caused by the same pathogen is possible because the biofilm on endotracheal tubes allows persistence and reintroduction of the microorganism. Studies of VAP caused by *Pseudomonas* show that approximately half of recurrent cases are caused by a new strain. Some studies

TABLE 126-8 Empirical Antibiotic Treatment of Hospital-Acquired and Ventilator-Associated Pneumonia

NO RISK FACTORS FOR RESISTANT GRAM-NEGATIVE PATHOGEN	RISK FACTORS FOR RESISTANT GRAM-NEGATIVE PATHOGEN* (CHOOSE ONE FROM EACH COLUMN)	
Piperacillin-tazobactam (4.5 g IV q6h)	Piperacillin-tazobactam (4.5 g IV q6h)	Amikacin (15–20 mg/kg IV q24h)
Cefepime (2 g IV q8h)	Cefepime (2 g IV q8h)	Gentamicin (5–7 mg/kg IV q24h)
Levofloxacin (750 mg IV q24h)	Ceftazidime (2 g IV q8h)	Tobramycin (5–7 mg/kg IV q24h)
	Imipenem (500 mg IV q6h)	Ciprofloxacin (400 mg IV q8h)
	Meropenem (1 g IV q8h)	Levofloxacin (750 mg IV q24h)
		Colistin (loading dose of 5 mg/kg IV followed by maintenance doses of 2.5 mg $\times$ [1.5 $\times$ CrCl + 30] IV q12h)
		Polymyxin B (2.5–3.0 mg/kg per day IV in 2 divided doses)
Risk Factors for MRSA^b (Add to above)		
Linezolid (600 mg IV q12h) or		
Adjusted-dose vancomycin (trough level, 15–20 mg/dL)		

*Prior antibiotic therapy, prior hospitalization, local antibiogram. ^bPrior antibiotic therapy, prior hospitalization, known MRSA colonization, chronic hemodialysis, local documented MRSA pneumonia rate $>10\%$ (or local rate unknown).

Abbreviations: CrCl, creatinine clearance rate; MRSA, methicillin-resistant *Staphylococcus aureus*.

have suggested that treatment failure may be less common with optimized β -lactam dosing and use of either prolonged or continuous infusion therapy.

Treatment failure and its cause are very difficult to determine early in the therapeutic course. Pneumonia due to superinfection, the presence of extrapulmonary infection, and drug toxicity must be considered. Serial quantitative cultures may clarify the microbiologic response, but biomarkers such as PCT are of uncertain value in this setting.

COMPLICATIONS

Apart from death, the major complication of VAP is prolongation of mechanical ventilation, with corresponding increases in the duration of ICU and hospital stay. In most studies, the common need for an additional week of mechanical ventilation resulting from VAP justifies aggressive efforts at prevention.

In rare cases, necrotizing pneumonia (e.g., due to *P. aeruginosa* or *S. aureus*) can cause significant pulmonary hemorrhage. More commonly, necrotizing infections result in the long-term complications of bronchiectasis and parenchymal scarring leading to recurrent pneumonia. Other long-term complications of pneumonia can include long-term oxygen therapy, a catabolic state in a patient already nutritionally at risk, the need for prolonged rehabilitation, and—in the elderly—an inability to return to independent function and the need for nursing home placement.

FOLLOW-UP

Clinical improvement, if it occurs, is usually evident within 48–72 h of the initiation of antimicrobial treatment, usually with an improvement in oxygenation. Because findings on chest radiography often worsen initially during treatment, they are less helpful than clinical criteria as an indicator of response to therapy.

PROGNOSIS

VAP is associated with crude mortality rates as high as 50–70%, but the real issue is attributable mortality. Many patients with VAP have underlying diseases that would result in death even if VAP did not occur. Attributable mortality exceeded 25% in one matched-cohort study, while more recent studies have suggested much lower rates. Some variability in VAP mortality rates is clearly related to the type of patient and ICU studied. VAP in trauma patients is not associated with attributable mortality, possibly because many of the patients were otherwise healthy before being injured. The causative pathogen also plays a major role. Generally, MDR pathogens are associated with significantly greater attributable mortality than non-MDR pathogens. Pneumonia caused by some pathogens (e.g., *S. maltophilia*) is simply a marker for a patient whose immune system is so compromised that death is almost inevitable.

PREVENTION (TABLE 126-7)

Because endotracheal intubation is a risk factor for VAP, the most important preventive intervention is to avoid intubation or minimize its duration. Successful noninvasive ventilation avoids many of the problems associated with endotracheal tubes. Strategies that minimize the duration of ventilation through daily holding of sedation and formal weaning protocols have also been highly effective in preventing VAP.

Unfortunately, a tradeoff in risks is sometimes necessary. Aggressive attempts to extubate early may result in reintubation(s) and increase aspiration, posing a risk of VAP. Heavy continuous sedation increases VAP risk, but self-extubation because of insufficient sedation is also a risk. The tradeoffs also apply to antibiotic therapy. Short-course antibiotic prophylaxis can decrease the risk of early-onset VAP in comatose patients requiring intubation, and data suggest that antibiotics decrease VAP rates overall. Conversely, prolonged courses of antibiotics consistently increase the risk of MDR VAP; pseudomonal VAP is rare among patients who have not recently received antibiotics.

Minimizing microaspiration around the endotracheal tube cuff also can prevent VAP. Simply elevating the head of the bed (at least 30° above horizontal, but preferably 45°) and using specially modified endotracheal tubes that allow removal of the secretions pooled above

the cuff can prevent microaspiration. The risk-to-benefit ratio of transporting the patient outside the ICU for diagnostic tests or procedures should be carefully considered since VAP rates are increased among transported patients.

The role played by overgrowth of the normal bowel flora in the stomach—in the presence of elevated gastric pH—in the pathogenesis of VAP is questionable. Therefore, avoidance of agents that raise gastric pH may be relevant only in certain populations, such as liver transplant recipients and patients who have undergone other major intraabdominal procedures or who have bowel obstruction. MRSA and nonfermenters such as *P. aeruginosa* and *Acinetobacter* species are not normally part of the bowel flora but reside primarily in the nose and on the skin, respectively.

In outbreaks of VAP due to specific pathogens, the possibility of a breakdown in infection control measures (particularly contamination of reusable equipment) should be investigated. Even high rates of pathogens that are already common in a particular ICU may result from cross-infection. Education and reminders of the need for consistent hand washing and other infection-control practices can minimize this risk.

HOSPITAL-ACQUIRED PNEUMONIA

While less well studied than VAP, HAP in non-intubated patients—both inside and outside the ICU—is similar to VAP. The main differences are the higher frequency of non-MDR pathogens and the generally better underlying host immunity in non-intubated patients. The lower frequency of MDR pathogens allows monotherapy in a larger proportion of cases of HAP than of VAP. However, the bacteriology and outcome of ventilated HAP patients may be very similar to those of patients with VAP.

The only pathogens that may be more common in the non-VAP population are anaerobes because of a greater risk of macroaspiration and the lower oxygen tensions in the lower respiratory tract of these patients. Anaerobes usually contribute only to polymicrobial pneumonias, and specific therapy targeting anaerobes probably is not needed since many of the recommended antibiotics are active against anaerobes.

Diagnosis is even more difficult for HAP in the non-intubated patient than for VAP. Lower respiratory tract samples appropriate for culture are considerably more difficult to obtain from non-intubated patients. Many of the underlying diseases that predispose a patient to HAP are also associated with an inability to cough adequately. Since blood cultures are infrequently positive (<15% of cases), the majority of patients with HAP do not have culture data on which antibiotic modifications can be based, and de-escalation is less likely. Despite these difficulties, the better host defenses in non-ICU patients result in lower mortality rates than are documented for VAP and for ventilated HAP. In addition, the risk of antibiotic failure is lower in HAP.

GLOBAL IMPACT

From the available data, it is virtually impossible to accurately assess the impact of pneumonia from a global perspective. Any differences in incidence, disease burden, and costs across different age, ethnic, and racial groups are compounded by differences among countries in terms of etiologic pathogens, resistance rates, access to health-care and diagnostic facilities, and vaccine availability and use.

A standard approach with clearly defined outcome measures is needed before the impact of pneumonia can be accurately evaluated. However, simple extrapolation from U.S. data for CAP and HAP/VAP shows that pneumonia has a significant impact on quality of life, morbidity, health costs, and mortality rates and that this impact has implications for patients and for society as a whole.

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FURTHER READING

CHASTRE J et al: Comparison of 8 vs 15 days of antibiotic therapy for ventilator-associated pneumonia in adults: A randomized trial. JAMA 290:2588, 2003.